

## Chain Rule Worksheet — Worked Answer Key

Twenty-four composition problems from basic powers through nested, product, and quotient combinations.

**Revision:** 2026-07-23    **Canonical page:** <https://bettergrades.net/subjects/math/calculus/worksheets/chain-rule/>

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1. Differentiate  $y = (3x + 2)^5$ .

**Answer:**  $y' = 15(3x + 2)^4$

- Use the power rule on the outer power  $u^5$ .
- Multiply by the inner derivative 3.

2. Differentiate  $y = (5x - 1)^4$ .

**Answer:**  $y' = 20(5x - 1)^3$

- Use the power rule on the outer power  $u^4$ .
- Multiply by the inner derivative 5.

3. Differentiate  $y = (2x + 4)^6$ .

**Answer:**  $y' = 12(2x + 4)^5$

- Use the power rule on the outer power  $u^6$ .
- Multiply by the inner derivative 2.

4. Differentiate  $y = (-3x + 2)^3$ .

**Answer:**  $y' = -9(-3x + 2)^2$

- Use the power rule on the outer power  $u^3$ .
- Multiply by the inner derivative  $-3$ .

5. Differentiate  $y = (7x + 1)^2$ .

**Answer:**  $y' = 14(7x + 1)^1$

- Use the power rule on the outer power  $u^2$ .
- Multiply by the inner derivative 7.

6. Differentiate  $y = (4x - 5)^3$ .

**Answer:**  $y' = 12(4x - 5)^2$

- Use the power rule on the outer power  $u^3$ .
- Multiply by the inner derivative 4.

7. Differentiate  $y = e^{3x^2}$ .

**Answer:**  $y' = 6xe^{3x^2}$

- Treat  $3x^2$  as the inner function.
- Differentiate the outer function and multiply by  $6x$ .

8. Differentiate  $y = e^{\sin x}$ .

**Answer:**  $y' = e^{\sin x} \cos x$

- Treat  $\sin x$  as the inner function.
- Differentiate the outer function and multiply by  $\cos x$ .

9. Differentiate  $y = \ln(5x - 2)$ .

**Answer:**  $y' = \frac{5}{5x-2}$

- Treat  $5x - 2$  as the inner function.
- Differentiate the outer function and multiply by 5.

10. Differentiate  $y = \ln(x^2 + 4)$ .

**Answer:**  $y' = \frac{2x}{x^2+4}$

- Treat  $x^2 + 4$  as the inner function.
- Differentiate the outer function and multiply by  $2x$ .

11. Differentiate  $y = \sin(4x^3)$ .

**Answer:**  $y' = 12x^2 \cos(4x^3)$

- Identify the outer trigonometric function and preserve its inner input.
- Differentiate the inner expression and multiply.

12. Differentiate  $y = \cos(2x - 1)$ .

**Answer:**  $y' = -2 \sin(2x - 1)$

- Identify the outer trigonometric function and preserve its inner input.
- Differentiate the inner expression and multiply.

13. Differentiate  $y = \tan(x^2)$ .

**Answer:**  $y' = 2x \sec^2(x^2)$

- Identify the outer trigonometric function and preserve its inner input.
- Differentiate the inner expression and multiply.

14. Differentiate  $y = \sin^3 x$ .

**Answer:**  $y' = 3 \sin^2 x \cos x$

- Identify the outer trigonometric function and preserve its inner input.
- Differentiate the inner expression and multiply.

15. Differentiate  $y = (1 + (2x - 3)^2)^4$ .

**Answer:**  $y' = 16(2x - 3)(1 + (2x - 3)^2)^3$

- Write the composition as nested layers from outside to inside.
- Differentiate each layer once and multiply the factors.

16. Differentiate  $y = e^{\sqrt{1+x^2}}$ .

**Answer:**  $y' = \frac{xe^{\sqrt{1+x^2}}}{\sqrt{1+x^2}}$

- Write the composition as nested layers from outside to inside.
- Differentiate each layer once and multiply the factors.

17. Differentiate  $y = \ln(1 + \sin^2 x)$ .

**Answer:**  $y' = \frac{2 \sin x \cos x}{1 + \sin^2 x}$

- Write the composition as nested layers from outside to inside.
- Differentiate each layer once and multiply the factors.

18. Differentiate  $y = x^2 e^{3x}$ .

**Answer:**  $y' = e^{3x}(2x + 3x^2)$

- Apply the product plus chain structure before simplifying.
- Retain every factor contributed by an inner derivative.
- A factored final form is  $e^{3x}(2x + 3x^2)$ .

19. Differentiate  $y = x \sin(x^2)$ .

**Answer:**  $y' = \sin(x^2) + 2x^2 \cos(x^2)$

- Apply the product plus chain structure before simplifying.
- Retain every factor contributed by an inner derivative.
- A factored final form is  $\sin(x^2) + 2x^2 \cos(x^2)$ .

20. Differentiate  $y = \frac{\ln x}{x^2}$ .

**Answer:**  $y' = \frac{1-2\ln x}{x^3}$

- Apply the quotient plus chain structure before simplifying.
- Retain every factor contributed by an inner derivative.
- A factored final form is  $\frac{1-2\ln x}{x^3}$ .

21. Differentiate  $y = \frac{e^x}{1+x^2}$ .

**Answer:**  $y' = \frac{e^x(1+x^2-2x)}{(1+x^2)^2}$

- Apply the quotient rule structure before simplifying.
- Retain every factor contributed by an inner derivative.
- A factored final form is  $\frac{e^x(1+x^2-2x)}{(1+x^2)^2}$ .

22. Differentiate  $y = (x^2 + 1)^3(x - 2)$ .

**Answer:**  $y' = (x^2 + 1)^2(7x^2 - 12x + 1)$

- Apply the product plus chain structure before simplifying.
- Retain every factor contributed by an inner derivative.
- A factored final form is  $(x^2 + 1)^2(7x^2 - 12x + 1)$ .

23. Differentiate  $y = \frac{\sin(2x)}{x}$ .

**Answer:**  $y' = \frac{2x \cos(2x) - \sin(2x)}{x^2}$

- Apply the quotient plus chain structure before simplifying.
- Retain every factor contributed by an inner derivative.
- A factored final form is  $\frac{2x \cos(2x) - \sin(2x)}{x^2}$ .

24. Differentiate  $y = \sqrt{1 + e^{2x}}$ .

**Answer:**  $y' = \frac{e^{2x}}{\sqrt{1+e^{2x}}}$

- Apply the nested chain structure before simplifying.
- Retain every factor contributed by an inner derivative.
- A factored final form is  $\frac{e^{2x}}{\sqrt{1+e^{2x}}}$ .